



Brock University
Faculty of Mathematics & Science
Department of Computer Science

COSC 3P93: Parallel Computing
4P78 Project Step 3

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1 Intro

2 Performance Testing

My Computer

	360x240	720x480	1080x1920
halo	2.863831261	7.661210509	30.610816487
bricks	15.252794018	22.988769295	60.991727457
wavy	2.615458843	9.626839493	47.173488550

Table 1: Computer 1, Single Threaded, No Disk

-halo720 7.661210509 -halo1080 30.610816487 -bricks720 22.988769295 -bricks1080 60.991727457
-wavy720 9.626839493 -wavy1080 47.173488550 -halo720-f 26.053255069 -halo1080-f 136.373871959
-bricks720-f 40.719280813 -bricks1080-f 170.722502377 -wavy720-f 28.109580179 -wavy1080-f
138.281518099 mp-halo720 3.558991764 mp-halo1080 14.919492215 mp-bricks720 10.571517149
mp-bricks1080 27.352880391 mp-wavy720 3.108623945 mp-wavy1080 15.890781591 mp-halo720-
f 23.085340934 mp-halo1080-f 119.350630179 mp-bricks720-f 28.311246929 mp-bricks1080-f
132.492900117 mp-wavy720-f 21.929993017 mp-wavy1080-f 104.948913426 mpi-halo720 5.155524278
mpi-halo1080 15.717249188 mpi-bricks720 13.163617292 mpi-bricks1080 30.210486661 mpi-
wavy720 5.338391755 mpi-wavy1080 21.866477872 mpi-halo720-f 12.120452874 mpi-halo1080-
f 58.323059184 mpi-bricks720-f 26.513881988 mpi-bricks1080-f 127.112395049 mpi-wavy720-f
20.710505775 mpi-wavy1080-f 100.060047423 mp-mpi-halo720 4.891824994 mp-mpi-halo1080
15.339569484 mp-mpi-bricks720 12.557625108 mp-mpi-bricks1080 28.491279222 mp-mpi-wavy720
4.679881252 mp-mpi-wavy1080 18.345591749 mp-mpi-halo720-f 12.982671860 mp-mpi-halo1080-
f 58.661721214 mp-mpi-bricks720-f 24.990997518 mp-mpi-bricks1080-f 126.449484168 mp-
mpi-wavy720-f 21.063672793 mp-mpi-wavy1080-f 99.319075376

-halo360 2.863831261 -bricks360 15.252794018 -wavy360 2.615458843 -halo360-f 8.099943463
-bricks360-f 19.848059572 -wavy360-f 7.501858413 mp-halo360 1.441681635 mp-bricks360
7.399861928 mp-wavy360 .685643623 mp-halo360-f 7.029366780 mp-bricks360-f 11.988864908
mp-wavy360-f 5.499844858 mpi-halo360 3.005354565 mpi-bricks360 10.288846674 mpi-wavy360
2.442392211 mpi-halo360-f 5.062579718 mpi-bricks360-f 12.089300147 mpi-wavy360-f 6.462228776
mp-mpi-halo360 2.874481385 mp-mpi-bricks360 9.780538278 mp-mpi-wavy360 2.193176283
mp-mpi-halo360-f 4.935568941 mp-mpi-bricks360-f 11.267476112 mp-mpi-wavy360-f 6.168513181

bretts computer

-halo360 3.651463603 -bricks360 21.515869878 -wavy360 3.425032398 -halo360-f 9.313003121
-bricks360-f 25.133392014 -wavy360-f 8.680104152 mp-halo360 1.521622857 mp-bricks360
8.277552774 mp-wavy360 .759378029 mp-halo360-f 9.333331259 mp-bricks360-f 13.901321742
mp-wavy360-f 7.446619034 mpi-halo360 2.277572908 mpi-bricks360 11.946162651 mpi-wavy360
1.666743752 mpi-halo360-f 4.344833085 mpi-bricks360-f 13.293646217 mpi-wavy360-f 5.973292092

mp-mpi-halo360 3.773099477 mp-mpi-bricks360 13.378941801 mp-mpi-wavy360 2.956704110
mp-mpi-halo360-f 9.984898870 mp-mpi-bricks360-f 19.850255344 mp-mpi-wavy360-f 10.108940873

References

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